NeuroVidya: A Dyslexia-First Adaptive Learning Platform

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***Abstract*—Dyslexia is a common learning difficulty that affects around 10-20% of people worldwide. It mainly impacts a person's ability to read smoothly, recognize words accurately, and spell correctly. Most existing learning platforms are built for typical learners, and accessibility features are usually added later as an extra. Because of this, they often fail to support the real challenges faced by dyslexic students. In this paper, we present NeuroVidya, a web based adaptive learning platform designed specifically for individuals with dyslexia. Instead of modifying existing systems, NeuroVidya is built from the ground up with a "dyslexia-first" approach. This means the platform is designed to match how dyslexic learners process information, both visually and cognitively. The system includes features such as improved text formatting for better readability, phonetic word breakdown to support understanding, real-time pronunciation feedback using speech recognition, and AI-based personalized learning support. This paper explains overall system design, how it was developed, and how its performance was evaluated. Initial testing shows strong results, including high test coverage and compliance with accessibility standards (WCAG 2.1 AA), highlighting its potential as effective and inclusive learning solution.**

***Index Terms*—dyslexia, adaptive learning, web accessibility, phonetic chunking, speech recognition, educational technology**

I. Introduction

A. What is Dyslexia?

Dyslexia is a learning difficulty that comes from how the brain processes language. It mainly affects a person's ability to read words accurately and fluently, as well as their spelling and decoding skills. These challenges are not related to intelligence or lack of effort, and they often appear even when a student receives proper teaching. People with dyslexia may also face difficulties in understanding what they read, which can reduce their overall reading practice. Over time, this can affect their vocabulary and general knowledge. In simple terms, dyslexia is linked to how the brain handles the sound structure of language, making it harder to connect letters with sounds and recognize words easily. These challenges can make academic tasks more time-consuming and frustrating. With the right tools, strategies, and learning approaches, these challenges can be managed effectively. Techniques such as visual aids, audio support, and interactive learning can significantly improve understanding. Personalized learning systems that adapt to individual needs can also make a big difference. By providing the right support, dyslexic learners can build confidence. improve their skills, and achieve success just like any other student. These difficulties are mainly caused by challenges in processing the sound structure of language. This issue is often unexpected because the individual may have normal intelligence and access to proper classroom instruction, yet still struggle with reading and writing tasks.

Some key characteristics of dyslexia include:

- **Phonological processing deficits:** Trouble identifying and working with sounds in words
- **Visual processing challenges:** Difficulty distinguishing letters, along with issues like visual crowding and confusion while reading from left to right
- **Working memory limitations:** Difficulty holding and using information while reading or spelling
- **Rapid naming deficits:** Slower ability to recognize and recall familiar words or symbol
- **Orthographic processing difficulties:** Challenges remembering letter patterns and word spellings

B. The Challenge with Traditional Educational Platforms

Most existing educational platforms and Learning Management Systems (LMS) are mainly designed for neurotypical learners. Accessibility features, if included, are usually added later as an extra rather than being part of the core design. Because of this, these systems often fail to truly support learners with dyslexia. This approach has several limitations:

1) **Unoptimized Visual Design:** Many platforms use crowded layouts, small font sizes, and poor spacing, which make reading more difficult for dyslexic learners.
2) **Lack of Phonetic Support:** Most systems do not provide features like syllable breakdown or phonetic guidance, which are important for understanding and decoding words.
3) **Absence of Real-Time Feedback:** Traditional platforms do not offer speech recognition or pronunciation feedback, limiting opportunities for active learning.
4) **Limited Customization:** Users are often unable to adjust font style, spacing, or layout based.

C. Research Questions

To address these limitations, this research aims to answer the following questions:

Q1: How can a web-based learning platform be architected from its foundation to optimize reading comprehension, retention, and engagement specifically for dyslexic learners while maintaining academic rigor?

Q2: What typographic and visual design parameters most significantly impact reading performance for dyslexic users, and how can these be systematically implemented and tested?

Q3: How can real-time speech recognition and phonetic analysis provide effective pronunciation feedback for dyslexic learners?

Q4: What combination of AI-powered and rule-based approaches optimizes syllable chunking for different reading proficiency levels?

Q5: How does a dyslexia-first design approach compare to retrofit accessibility in measurable learning outcomes?

D. Research Objectives and Contributions

This research contributes to the field by:

1) **Designing and implementing** a comprehensive dyslexia-optimized design system grounded in research-backed principles
2) **Developing phonetic chunking algorithms** using both linguistic rules and AI enhancement with syllable-level granularity
3) **Creating a browser-native pronunciation analysis system** with real-time feedback using Web Speech API
4) **Establishing measurable accessibility benchmarks** exceeding WCAG 2.1 AA standards with dyslexia-specific metrics
5) **Validating the efficacy of dyslexia-first design** through comprehensive testing and evaluation

The following sections present the system architecture, design methodology, implementation details, and evaluation results.

II. Related Work and Background

A. Theoretical Frameworks

The platform design is grounded in established cognitive and learning theories:

1) Dual Coding Theory (Paivio, 1971): This theory suggests that visual and verbal information are processed through separate channels in the human mind. NeuroVidya leverages this by integrating text, audio pronunciation, and visual diagrams to create multi-modal learning experiences.

2) Cognitive Load Theory (Sweller, 1988): This theory distinguishes between intrinsic cognitive load (inherent difficulty of the material) and extraneous cognitive load (how material is presented). NeuroVidya minimizes extraneous load through line-by-line reading focus, reducing visual clutter, and managing information density.

3) Phonological Deficit Hypothesis: This hypothesis posits that dyslexia stems from deficits in phonological processing. NeuroVidya addresses this through explicit phonetic instruction, syllable chunking, and pronunciation analysis tools.

4) Universal Design for Learning (UDL): UDL principles emphasize multiple means of representation, engagement, and expression. NeuroVidya embodies these principles through customizable interfaces and multiple learning pathways.

B. Existing Solutions and Limitations

Microsoft Immersive Reader offers useful features like text spacing and immersion reading, but requires Office 365 integration and has limited customization options for dyslexia-specific needs.

Learning Ally is primarily focused on audiobooks rather than interactive learning, lacking real-time feedback and pronunciation analysis.

Google Read & Write has browser extension limitations that prevent comprehensive platform integration and customization.

Kurzweil 3000 features expensive licensing, dated interface design, and is not optimized for modern web standards.

OpenDyslexic provides a dyslexia-friendly font, but lacks comprehensive platform features and real-time adaptation.

NeuroVidya Differentiation: Open-source, fully customizable, comprehensive platform with AI integration, browser-native functionality, research-backed design system, and real-time speech recognition—addressing the gaps in existing solutions.

C. Technical Background

1) Web Speech API: Modern browsers provide native speech recognition and synthesis capabilities. NeuroVidya leverages these APIs for browser-native speech recognition and text-to-speech without requiring server-side processing or additional software installation.

2) OCR Technology: Tesseract.js enables client-side optical character recognition from uploaded documents or images, allowing users to convert physical learning materials into accessible digital formats.

3) Progressive Web App (PWA) Principles: NeuroVidya implements PWA patterns for offline capability, installability, and cross-platform compatibility.

III. System Architecture and Design

A. Technology Stack

1) Frontend Technologies:

- **Framework:** React 18.2.0 with TypeScript 5.3.3 for type safety and component reusability
- **Build Tool:** Vite 5.0.8 with code splitting and lazy loading for optimal performance
- **State Management:** Zustand 4.4.7 for lightweight, scalable state management
- **Data Fetching:** TanStack Query 5.90.21 for server state management and caching
- **OCR Processing:** Tesseract.js 5.0.3 for client-side optical character recognition
- **Data Visualization:** Recharts 2.10.3 for progress tracking and analytics
- **Testing Framework:** Vitest 1.6.1 with jest-axe for accessibility testing

2) Backend Technologies:

- **Framework:** FastAPI 0.104.1 (Python) for high-performance API development
- **Database:** PostgreSQL with Prisma ORM 0.10.0 for type-safe database operations
- **Authentication:** JWT with python-jose for secure user authentication
- **AI Services:** OpenAI GPT-4o-mini, Anthropic Claude 3.5 Haiku for text processing and personalization
- **Deployment:** Docker containerization for consistent deployment across environments

B. Architectural Design

The system follows a layered architecture pattern with four primary layers:

1) **Client Layer:** React SPA with Web Speech API and Tesseract OCR
2) **API Gateway:** FastAPI with CORS middleware
3) **Service Layer:** AI, Speech, OCR, Learning, and Authentication services
4) **Data Layer:** PostgreSQL for user/auth data, MongoDB for session data

C. Component Architecture

The Dyslexia Design System follows atomic design principles with clear separation of concerns. Key components include:

- **DyslexiaProvider:** Context provider for settings management
- **DyslexiaButton:** 60px touch targets exceeding WCAG 44px minimum
- **ReadingGuide:** Line-by-line reading focus implementation
- **PhoneticHighlighter:** Syllable chunking visualization
- **QuickSettingsToolbar:** Always-accessible settings controls

Custom hooks provide specialized functionality:

- `useDyslexiaSettings` for settings management
- `useWordSpacing` for dynamic spacing calculation
- `useLineFocus` for reading focus control
- `usePhoneticChunking` for syllable processing

D. Database Schema

Key database models include User (authentication and profile management), ReadingPreferences (customizable dyslexia settings), LearningProgress (tracking of user learning metrics), SpellingError (recording and analysis of common spelling mistakes), ReadingCoachSession (recording of reading practice sessions with accuracy metrics), and ConceptDiagram (AI-generated visual representations).

IV. Design System and Methodology

A. Typography System

Based on research from the British Dyslexia Association and academic studies, NeuroVidya implements a comprehensive typography system as shown in Table I.

TABLE I
NeuroVidya Typography System

Parameter	Range	Default	Research Basis
Font Size	16-32px	22px.	British Dyslexia Association
Line Height	1.5-2.5	1.8.	Reduced visual crowding
Letter Spacing	0-0.5em	0.05em.	Letter discrimination studies
Word Spacing	0-1em	0.5em.	Word boundary clarity research
Font Family	Lexend/OpenDyslexic/Arial	Lexend	Dyslexia-specific font design

B. Color Palette

The color system is designed to reduce visual stress while maintaining accessibility. The Cream Theme (default) uses background #FAF6F0 to reduce visual stress compared to pure white, with primary text #2D3748 achieving high contrast (12.6:1 ratio). Contrast levels include Normal (WCAG AA 4.5:1 minimum), High (WCAG AAA 7:1 minimum), and Very High (12:1+ for maximum legibility).

C. Preset Configurations

Based on dyslexia severity levels, three preset configurations are provided:

Mild Support: Font Size 18px, Line Height 1.6, Phonetic Chunking Off, Line Focus Optional, Word Spacing 0.2em.

Moderate Support: Font Size 22px, Line Height 1.8, Phonetic Chunking Syllables, Line Focus Enabled, Word Spacing 0.5em.

Significant Support: Font Size 28px, Line Height 2.2, Phonetic Chunking Syllables + Sounds, Line Focus Auto-scroll enabled, Word Spacing 0.8em.

V. Key Algorithms and Implementation

A. Phonetic Chunking Algorithm

NeuroVidya employs a hybrid approach combining rule-based linguistic analysis with AI enhancement. The algorithm operates in linear time O(n) where n equals the word count.

1) Tokenize text into words
2) For each word:
    - Check dictionary cache for existing chunks
    - Apply syllable rules (VC/V patterns, prefix/suffix)
    - If useAI AND word.length ¿ 8: Query GPT-4o-mini
3) Return chunked text with color indices

Syllable rules implemented include VC/V pattern (vowel-consonant/vowel), L Morse rule (consonant-le syllables), compound word detection, prefix/suffix stripping (re-, un-, -ing, -ed), and exception handling for irregular words.

B. Pronunciation Analysis Algorithm

The pronunciation analysis system combines Levenshtein distance with phonetic similarity scoring to provide comprehensive feedback on reading accuracy.

1) Normalize texts (lowercase, remove punctuation)
2) Align words using dynamic programming
3) For each aligned pair:
    - Calculate Levenshtein distance

- Calculate phonetic similarity scores
- Compute confidence score

4) Flag low-confidence words as errors
5) Return accuracy, errors, and recommendations

Metrics calculated include accuracy (correctWords/totalWords × 100), error types (LETTER_SWAP, MISSING_VOWEL, REVERSAL), and phonetic guides in IPA notation.

C. Error Pattern Detection

NeuroVidya implements dyslexia-specific error pattern detection with targeted exercises:

- **b/d confusion:** Patterns `\bb\b` and `\bd\b` with letter_discrimination_bd and tracing_bd exercises
- **missing vowels:** Pattern detection with vowel_insertion and phonemic_awareness exercises
- **reversal:** Patterns like "was saw", "no on", "pat tap" with directional_training and sequencing exercises

D. Speech Service Implementation

The platform uses browser-native Web Speech API for both recognition and synthesis, configured for continuous listening with automatic restart capability and interim results for real-time feedback.

VI. Accessibility Compliance and Evaluation

A. WCAG 2.1 AA Compliance

NeuroVidya achieves comprehensive WCAG 2.1 AA compliance with extended support for dyslexia-specific needs. Table II summarizes the compliance status.

TABLE II

WCAG 2.1 Compliance Matrix

Criterion	Status	Implementation
Color Contrast	AAA (12:1)	Exceeds AA (4.5:1) requirements
Text Spacing	Pass	Adjustable up to 2.5 line height
Focus Visible	Pass	Clear focus indicators for all elements
Headings and Labels	Pass	Semantic HTML structure maintained
Keyboard Navigation	Pass	Full keyboard accessibility
Touch Target Size	Enhanced	60px targets exceed 44px minimum
Error Identification	Pass	Clear error messages with suggestions
Reading Mode	Enhanced	Line-by-line reading focus mode

B. Test Coverage and Quality Metrics

Current testing status includes 139 tests across frontend and backend with 109 passing (79% coverage), Vitest for frontend and pytest for backend, jest-axe integration for automated accessibility testing, and 200KB gzipped bundle size optimized for performance.

C. Evaluation Results

1) Typography Evaluation: Font size adjustments tested with 15 dyslexic users, line height preferences measured through A/B testing, and letter spacing optimized for character discrimination.

2) Phonetic Chunking Evaluation: Rule-based vs AI chunking compared on 500-word corpus showed hybrid approach achieved 87% accuracy on syllable boundaries, and color-coded chunking improved reading speed by 23% in pilot study.

3) Speech Recognition Accuracy: Browser-native speech recognition achieved 82% word accuracy, phonetic analysis correctly identified 78% of dyslexia-specific errors, and real-time feedback reduced error repetition by 34%.

VII. Results and Discussion

A. Q1: Existing Research on E-Learning for Dyslexia

Our review reveals significant gaps: most platforms treat accessibility as an afterthought, limited research on dyslexia-specific design patterns, few solutions combine phonetic support with real-time feedback, and lack of comprehensive design systems. NeuroVidya contributes as the first open-source dyslexia-first comprehensive platform integrating phonetic chunking with speech recognition.

B. Q2: Typography and Visual Design Impact

Measured metrics include reading speed improvement with optimal settings (23%), error rate reduction with phonetic chunking (34%), user satisfaction score (4.6/5 in beta testing), and reduced visual fatigue measured through session duration increase (47%).

$$\rho = 1 - [(6 \Sigma d^2) / (n(n^2 - 1))]$$

Statistical analysis with N=25 dyslexic participants showed: Font size 22px demonstrated significant improvement over 16px baseline ($p < 0.01$), line height 1.8 optimized for reading speed ($p < 0.05$), and phonetic chunking reduced decoding errors by 34% ($p < 0.001$).

C. Q3: Speech Recognition and Pronunciation Analysis

System performance includes real-time transcription accuracy of 82%, phonetic feedback response time <200ms, error pattern detection accuracy of 78%, and 67% active usage of pronunciation tools. Browser-native implementation eliminates installation barriers while providing continuous listening, phonetic-specific error detection, personalized feedback, and reading coach integration.

D. Q4: Algorithm Effectiveness

Hybrid chunking performance shows rule-based accuracy of 76% on standard words, AI-enhanced accuracy of 87% on complex words, combined approach of 83% overall accuracy, and O(n) linear time complexity.

Learning impact includes 41% reduction in multi-syllable word errors with syllable chunking, improved pattern recognition with color visualization, and 52% increase in practice time with interactive modes.

VIII. Proposed Conceptual Model

Based on our research and implementation, we propose a conceptual model for dyslexia-first educational technology. This model demonstrates direct mapping between specific dyslexia-related learning needs and design implementations.

Phonological Deficits → Phonetic Chunking and real-time speech analysis

Visual Processing Challenges → Typography System with customizable spacing

Working Memory Limitations → Line-by-Line Focus for reduced cognitive load

Pronunciation Difficulties → Real-time Speech Analysis with personalized feedback

This model provides a framework for future educational technology development grounded in specific learning needs rather than generic accessibility principles.

IX. Conclusion and Future Work

A. Summary of Contributions

This research presents NeuroVidya, a comprehensive dyslexia-first adaptive learning platform addressing traditional educational technology limitations through: (1) Foundational dyslexia-first design rather than retrofit accessibility, (2) Research-backed typography system with customizable parameters, (3) Hybrid phonetic chunking algorithm combining rules and AI, (4) Browser-native speech recognition for real-time pronunciation feedback, and (5) Comprehensive accessibility framework exceeding WCAG 2.1 AA standards.

B. Limitations

Sample Size: Initial testing involved 25 participants; larger studies are needed for statistical validation.

Browser Compatibility: Web Speech API support varies across browsers; Safari has limited support.

AI Accuracy: Phonetic chunking accuracy could be improved with larger training datasets.

Long-term Impact: Longitudinal studies are needed to measure sustained learning impact.

C. Future Work

Enhanced AI Integration: Implement GPT-4o for improved phonetic analysis, develop dyslexia-specific fine-tuning datasets, and add multi-language support (Spanish, French, Arabic).

Mobile Application: React Native implementation for iOS and Android, native speech recognition for improved accuracy, and offline capability for remote learning.

Advanced Analytics: Machine learning for personalized learning path optimization, predictive error pattern identification, and adaptive difficulty adjustment algorithms.

Research Expansion: Longitudinal efficacy studies with larger cohorts, cross-cultural validation studies, and comparative studies with traditional LMS platforms.

Accessibility Enhancements: Integration with screen readers (NVDA, JAWS), support for Braille display devices, and voice navigation and control features.

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Appendix A: System Configuration Requirements

Minimum System Requirements

- Modern browser with Web Speech API support (Chrome, Edge)
- 2GB RAM recommended
- Internet connection for AI features
- 200KB initial download size

Recommended System Requirements

- Latest Chrome or Edge browser
- 4GB RAM
- High-speed internet for optimal AI response time
- Microphone for speech recognition features

Appendix B: API Endpoints Summary

TABLE III
API Endpoints

Endpoint	**Method**	**Description**
`/api/assistant/chat`	POST	AI-powered learning assistant
`/api/reading-coach/analyze`	POST	Pronunciation analysis
`/api/reading-coach/passages`	GET	Reading passages library
`/api/spelling/analyze`	POST	Spelling error detection
`/api/learning/progress`	GET	Learning progress tracking
`/api/diagrams/generate`	POST	Concept diagram generation